Lecture 21: Root of Characteristic Polynomial of AR Models and Moving Average Models

1 Interpreting Roots of Characteristic Polynomial

Let us consider an AR(2) model. Recall the Yule-Walker equation $\gamma_l = \sum_{i=1}^{p} \phi_i \gamma_{l-i}$. The moment equation of a stationary AR(2) model is given by

$$\rho_l = \phi_1 \rho_{l-1} + \phi_2 \rho_{l-2}, \qquad l \geq 1.$$

For a stationary AR(2) series, one gets $\rho_1 = \frac{\phi_1}{1-\phi_2}$, and $\rho_l = \phi_1 \rho_{l-1} + \phi_2 \rho_{l-2}$ for $l \geq 2$. Hence, the autocorrelation function (ACF) of a stationary AR(2) series satisfies the second order differential equation

$$(1 - \phi_1 B - \phi_2 B^2)\rho_l = 0, \tag{1}$$

where B is called the *back-shift* operator, and defined as $B\rho_l = \rho_{l-1}$. This difference equation determines the properties of the ACF. Consider the second-order polynomial equation above, i.e., $1 - \phi_1 x - \phi_2 x^2 = 0$. The solutions of this equation are

$$x = \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{-2\phi_2}.$$

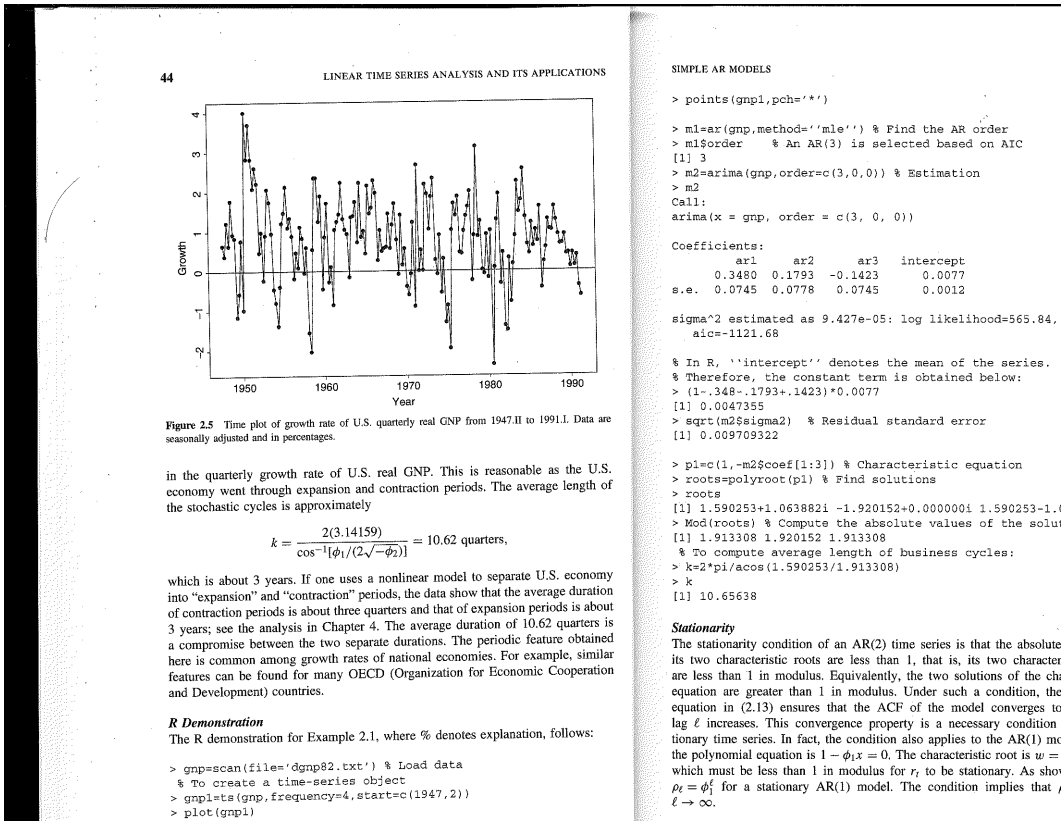
In the literature on time series, the reciprocals of the two solutions above are called the characteristic roots of the AR(2) model. Let ω_1 and ω_2 be the two characteristic roots (notice that $\omega_1 = \omega_2$ if $\phi_1^2 + 4\phi_2 = 0$). If they are both real valued, then the second-order difference equation of the model (1) can be factored as $(1 - \omega_1 B)(1 - \omega_2 B)$, where ω_1 and ω_2 are real. Then, this can be regarded as an AR(1) model that operates on top of another AR(1) model.

If $\phi_1^2 + 4\phi_2 < 0$, then we would have that ω_1 and ω_2 are complex numbers (a complex conjugate pair). The plot of the ACF of the return series r_t would then show a picture of damping sine and cosine waves. In financial applications, complex roots are important. They indicate business cycles. By writing the complex solutions $x = a \pm bi$, where $i = \sqrt{-1}$ is the imaginary unit, we can characterize the average length of the business cycle: this is given by the value k such that $2\pi/k = \theta$, where θ is the angle between $a + bi$ and the x-axis, so that in k steps, the angle goes back to 2π, i.e., we get a cycle. In other words, θ is the angle between (a, b) and $(1, 0)$. Notice that the cosine of this angle is given by the inner product of the corresponding unit norm vectors, i.e., $\langle \frac{1}{\sqrt{a^2+b^2}}(a, b), (1, 0) \rangle = \frac{a}{\sqrt{a^2+b^2}}$. Hence, $\theta = \cos^{-1}\left(a/(\sqrt{a^2 + b^2})\right)$, and as a result, $k = \frac{2\pi}{\cos^{-1}\left(a/(\sqrt{a^2+b^2})\right)}$, where the $\cos^{-1}$ is stated in radians. In our specific example where $x =$

$\frac{\phi_1 \pm \sqrt{\phi_1^2+4\phi_2}}{-2\phi_2}$, we have that[1]

$$k = \frac{2\pi}{\cos^{-1}\left(\phi_1/(2\sqrt{-\phi_2})\right)}. \tag{2}$$

Example: Consider the quarterly growth rate of the U.S. gross national product (GNP), seasonally adjusted[2], from the second quarter of 1947 to the first quarter of 1991. We report the time plot in Figure 1.



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Figure 2.5 Time plot of growth rate of U.S. quarterly real GNP from 1947.II to 1991.I. Data are seasonally adjusted and in percentages.

in the quarterly growth rate of U.S. real GNP. This is reasonable as the U.S. economy went through expansion and contraction periods. The average length of the stochastic cycles is approximately

$$k = \frac{2(3.14159)}{\cos^{-1}[\phi_1/(2\sqrt{-\phi_2})]} = 10.62 \text{ quarters},$$

which is about 3 years. If one uses a nonlinear model to separate U.S. economy into "expansion" and "contraction" periods, the data show that the average duration of contraction periods is about three quarters and that of expansion periods is about 3 years; see the analysis in Chapter 4. The average duration of 10.62 quarters is a compromise between the two separate durations. The periodic feature obtained here is common among growth rates of national economies. For example, similar features can be found for many OECD (Organization for Economic Cooperation and Development) countries.

R Demonstration

The R demonstration for Example 2.1, where % denotes explanation, follows:

```
> gnp=scan(file='dgnp82.txt') % Load data
 % To create a time-series object
> gnp1=ts(gnp,frequency=4,start=c(1947,2))
> plot(gnp1)
```

SIMPLE AR MODELS

```
> points(gnp1,pch='*')

> m1=ar(gnp,method=''mle'') % Find the AR order
> m1$order    % An AR(3) is selected based on AIC
[1] 3
> m2=arima(gnp,order=c(3,0,0)) % Estimation
> m2
Call:
arima(x = gnp, order = c(3, 0, 0))

Coefficients:
         ar1     ar2      ar3   intercept
      0.3480  0.1793  -0.1423      0.0077
s.e.  0.0745  0.0778   0.0745      0.0012

sigma^2 estimated as 9.427e-05: log likelihood=565.84,
   aic=-1121.68

% In R, ''intercept'' denotes the mean of the series.
% Therefore, the constant term is obtained below:
> (1-.348-.1793+.1423)*0.0077
[1] 0.0047355
> sqrt(m2$sigma2)  % Residual standard error
[1] 0.009709322

> p1=c(1,-m2$coef[1:3]) % Characteristic equation
> roots=polyroot(p1) % Find solutions
> roots
[1] 1.590253+1.063882i -1.920152+0.000000i 1.590253-1.(
> Mod(roots) % Compute the absolute values of the solut
[1] 1.913308 1.920152 1.913308
 % To compute average length of business cycles:
> k=2*pi/acos(1.590253/1.913308)
> k
[1] 10.65638
```

Stationarity

The stationarity condition of an AR(2) time series is that the absolute its two characteristic roots are less than 1, that is, its two character are less than 1 in modulus. Equivalently, the two solutions of the cha equation are greater than 1 in modulus. Under such a condition, the equation in (2.13) ensures that the ACF of the model converges to lag ℓ increases. This convergence property is a necessary condition tionary time series. In fact, the condition also applies to the AR(1) mo the polynomial equation is $1 - \phi_1 x = 0$, The characteristic root is $w =$ which must be less than 1 in modulus for r_t to be stationary. As sho $\rho_\ell = \phi_1^\ell$ for a stationary AR(1) model. The condition implies that $\ell \to \infty$.

Next, let us fit an AR(3) model to the data. Suppose we have already estimated the model, and found that the fitted model is given by

$$r_t = 0.0047 + 0.348 r_{t-1} + 0.179 r_{t-2} - 0.142 r_{t-3} + a_t,$$

and $\hat{\sigma}_a = 0.0097$. The ACF of this AR(3) satisfies the third-order difference equation

$$1 - 0.348B - 0.179B^2 + 0.142B^3 = 0,$$

[1]Observe that $\phi_2 < 0$, thus $\sqrt{\phi_2^2} = -\phi_2$.

[2]Seasonally adjusted refers to a process of removing the influence of regular, periodic fluctuations (like those related to seasons) from a time series data.

which can be factored as

$$(1 + 0.521B)(1 - 0.869B + 0.274B^2) = 0.$$

This representation captures the important features of the time series. The first factor shows the exponentially decaying feature of the GNP. Next, let us focus on the second-order factor $1-0.869B-(-0.274)B^2$. We deduce that $\phi_1^2 + 4\phi_2 = 0.869^2 + 4(-0.274) = -0.341 < 0$. Hence, the roots of the polynomial are complex, which confirms the existence of business cycles in the quarterly growth rate of U.S. GNP. This is in line with economic intuition, considering that the U.S. economy went through contraction and expansion periods. We can also compute the average length of the cycle k using Eq. (2). This leads to $k = \frac{2(3.14159)}{\cos^{-1}\left(\phi_1/(2\sqrt{-\phi_2})\right)} = 10.62$ quarters, which is roughly three years.

2 Moving Average Models

Moving average (MA) models is another class of tractable models that are useful for modeling return series in finance. They are useful to predict future values of a signal using the weighted moving average of past prediction errors, similar to the regression. We will introduce MA models as an infinite AR model, with some parameter constraints. Consider an AR model with infinite order as

$$r_t = \phi_0 + \phi_1 r_{t-1} + \phi_2 r_{t-2} + \ldots + a_t.$$

Such an AR model is not realistic because it has infinitely many parameters. To reduce the dimensionality and make the model practical, we can assume that the coefficients ϕ_i's satisfy some constraints so that they are determined by a finite number of parameters. For example,

$$r_t = \phi_0 - \theta_1 r_{t-1} - \theta_1^2 r_{t-2} - \theta_1^3 r_{t-3} + \ldots a_t, \tag{3}$$

where we have effectively imposed that $\phi_i = -\theta_1^i$. We require $|\theta_1| < 1$, otherwise the series will explode and the time series cannot be stationary. This, in turn, implies that $\theta_1^i \to 0$ as $i \to \infty$. Hence, the influence of r_{t-i} on r_t decays exponentially as i increases. We may rewrite Eq. (3) as

$$r_t + \theta_1 r_{t-1} + \theta_1^2 r_{t-2} + \theta_1^3 r_{t-3} + \cdots = \phi_0 + a_t. \tag{4}$$

This means that the model for r_{t-1} is

$$r_{t-1} + \theta_1 r_{t-2} + \theta_1^2 r_{t-3} + \cdots = \phi_0 + a_{t-1}. \tag{5}$$

Multiplying Eq. (5) by θ_1, and then subtracting it from Eq. (4), we obtain

$$r_t = \phi_0(1 - \theta_1) + a_t - \theta_1 a_{t-1}. \tag{6}$$

This means that, apart from the constant term $\phi_0(1 - \theta_1)$, r_t is a weighted average of the shocks a_t and a_{t-1}. For this reason, the model is called a MA model of order 1, or MA(1) for short.